



Autonomous Drone Racing

Summer Semester 2026 Course Syllabus

Course Overview

In this course, teams of up to two students will work on a joint project to program an advanced control algorithm for autonomous drone racing. The main goal of this course is to teach robotics problem-solving skills tailored to the high-speed, agile navigation of unmanned aerial vehicles (UAVs) through complex 3D tracks, as well as project management and teamwork.

Learning Objectives

You will learn to independently solve robotics problems specific to agile flight. This includes analyzing the challenge, developing a project plan, and trading off different solution approaches. Core learning outcomes emphasize mastering **simulation** to design, evaluate, and iterate on drone flight algorithms within a physics-based virtual environment. From there, you will explore various algorithmic methodologies, choosing to implement **model-based control** for precise dynamic tracking, leveraging **learning**-based methods like reinforcement learning, or exploring a hybrid of approaches. You will also develop practical skills in handling physical **hardware**, tackling the constraints of embedded systems, sensor noise, and actuation limits. Crucially, you will navigate **sim2real transfer**, bridging the reality gap to successfully deploy your simulated policies and controllers directly onto the physical racing drones. By developing an idea into a real-robot demonstration, you will gain hands-on implementation experience in fast-paced robot decision-making algorithms. You will also learn to work efficiently as a team, document your approach, and communicate your results. The skills acquired in this course are preparation for research, entrepreneurship, or a leadership position in the autonomous systems industry.

Prerequisites

The prerequisites for this course are

- fundamental knowledge in robotics, machine learning, and control theory,
- experience in Python or C++ programming and Linux (Ubuntu), and
- (*optional*) experience in the Robot Operating System (ROS).

Teaching Team

- **Instructor:** Prof. Dr. Angela Schoellig
- **Teaching assistants:** Martin Schuck, Marcel Rath, Radu Cristian, Yuming Li

Chair of Safety, Performance, and Reliability for Learning Systems (LSY)

School of Computation, Information and Technology, Technical University of Munich

Website: <https://www.learnsyslab.org>

Contact Email

If you have questions related to the course content or logistics, please send an email to teaching.lsy@xcit.tum.de and include "ADR Project" in the subject line.



Teaching and Learning Approach

Each team will work on their own control strategy for the drone race.

Lectures

Introductory lectures will be provided to teach approaches and tools for successful robotics problem-solving. These lectures are mandatory for all teams. Teams will be given relevant literature on drone racing for further self-study, and teaching assistants will support the teams with refining their project plan and evaluating different solution approaches.

Videos

In addition to the lectures, we will provide videos on specific topics in Moodle. These videos are meant for self-study and can be watched at each team's convenience.

Drone Lab Sessions

After the introductory lectures, there will be an introductory session on the hardware in our drone lab. Following the introduction, we will have lab sessions each week on the same day as the lecture where students can test their approach on the real drones. Students will sign up for these sessions a week in advance and get assigned time slots over the whole day. A teaching assistant will help you with your hardware questions. Students are not required to attend these sessions, but it is **highly** encouraged. Getting a reliable solution running on real robots usually takes time.

Online Competition

During the semester, we will run an online competition among teams that will automatically update your position based on your best-performing approach. You can find the current leaderboard on the [competition branch on Github](#). Teams will **not** be graded based on their position during the semester, but it is required to be considered for the course, see next section, and it will give you a sense of where you are in terms of performance and robustness compared to other teams.

Enrollment and Admission

To participate in the course, students must first register via TUMonline. Please note that it is standard procedure for all students to initially be placed on the waiting list upon registration.

As an initial qualification challenge, you will be tasked with programming a controller that flies a drone through a set of gates. To be **considered** for a spot in the course, teams must register via the designated Google Form so their times are properly submitted, and successfully complete this challenge, **regardless of the speed achieved**, by April 29th.

If the number of succeeding teams exceeds the maximum capacity of the course, admission will be granted to the fastest teams based on their performance in the qualification challenge.

Course Administration

We will use [Moodle](#) for course announcements and submission of required deliverables.

Schedule

All lectures will take place in N2408 in the N4 building. In the table, texts in **red** indicate mandatory items, and texts in *italic* indicate optional, but highly recommended items. On the 02.07.2026, an additional demo day is held, where you can present your real-life results to the



instructors and fellow students. Afterwards, there will be the LSY open lab day with demo sessions, announcements of open positions, and a get-together.

Lectures / Meeting Times (Wednesdays @ 13:15 – 14:45 CEST)		Deadlines and Action Items (Specified Dates @ 23:59 CEST)	
1	15.04.2026 Kickoff meeting		
2	22.04.2026 Project intro, installation, online competition		
3	29.04.2026 <i>Office hour</i> (13:15-14:15 in N0402)	29.04.2026	Submit to online challenge
4	06.05.2026 Selected algorithms for racing		
5	13.05.2026 Hardware, code, tools + Lab tour		
6	20.05.2026 <i>Lab session</i>		
7	27.05.2026 <i>Lab session</i>		
8	03.06.2026 <i>Lab session</i>		
9	10.06.2026 Project progress presentations	09.06.2026	Submit project progress presentation slides
10	17.06.2026 <i>Lab session</i>		
11	24.06.2026 <i>Lab session</i>		
12	01.07.2026 <i>Lab session</i>		
13	08.07.2026 Final presentations	07.07.2026	Submit report, slides, code

Course Attendance

It is required that the full team attends and presents at the two mandatory team presentation sessions. In addition, all team members are expected to attend the scheduled project meetings. Unexcused absences will lead to failure of the course. In case of illness, a medical certificate must be presented. Conflicts with other courses are not a sufficient excuse.

Project Resources

If necessary, teams will be asked to fill out forms to keep track of rented equipment and keys. After the completion of the course, any rented equipment and keys must be returned, and the return must be confirmed by the advisor.

Evaluation

For marking, we consider

- the quality of the implementation and the robustness and speed in the final demonstration (40%),
- a written report summarizing the results (30%, length of the report: up to 3 double-column pages per team member), and
- a final project presentation followed by a Q&A session (30%, length of presentation: 5



minutes per team member).

Final Report Guidelines

Your final written report must be formatted using the standard [IEEE conference template](#), which is also available directly on Overleaf. The main body of the document is **strictly limited to a maximum of 6 pages** (excluding references).

Proper academic citation is a fundamental requirement of this report. Citing your sources is crucial because it gives necessary credit to original authors, clearly distinguishes your own novel contributions from prior research, and provides a verifiable trail of evidence for your claims. If you are unsure how to format your references, please review the [IEEE Citation Guidelines](#).

Academic Integrity and AI Usage

You must ensure the absolute academic integrity of your submission. While you may use AI writing assistants to refine your language, **you are strictly and solely responsible for all content** within your report. This includes the validity of the ideas presented, ensuring that text is not plagiarized, and verifying that no hallucinated references are included.

If we detect uncredited text copies, hallucinated references, or any other forms of academic misconduct, it will result in an automatic **fail grade** for the entire course. To help avoid unintentional plagiarism, we recommend utilizing tools such as [Grammarly's Plagiarism Checker](#). Please note that while such tools are not perfect, they offer a helpful preliminary check before your final submission.